ChemComm

Chemical Communications

Accepted Manuscript

View Article Online
DOI: 10.1039/C9CC09990E

COMMUNICATION

Ammonia-assisted fast Li-ion conductivity in a new hemiammine lithium borohydride, $LiBH_4·½NH_3$

Received 00th January 20xx,
Accepted 00th January 20xx

DOI: 10.1039/x0xx00000x

Yigang Yan,[a,b] Jakob B. Grinderslev,[a] Young-Su Lee,[c] Mathias Jørgensen,[a] Young Whan Cho,[c] Radovan Černý,[d] Torben R. Jensen[a]*

Hemiammine lithium borohydride, $LiBH_4·½NH_3$, is characterized and a new Li^+ conductivity mechanism is identified. It exhibits a Li^+ conductivity of 7×10^{-4} S·cm^{-1} at 40 °C in the solid state and of 3.0×10^{-2} S·cm^{-1} at 55 °C after melting. The molten state of $LiBH_4·½NH_3$ has a high viscosity and can be mechanically stabilized in nano-composites with inert metal oxides and other hydrides making it a promising battery electrolyte.

Lithium-ion batteries enabled the success of small-size portable electronics and are also used to power tools and cars. Today, Li-ion batteries dominate the global market of rechargeable batteries. However, the energy capacity and power of Li-ion batteries are only incrementally improved due to intrinsic limitations. The organic liquid electrolytes used in these cells cause safety concerns due to flammability, limited thermal stability, along with incompatibility with lithium metal anodes and polarization effects, which limits the maximum cell current. These draw-backs may be eliminated by the discovery of solid inorganic electrolytes that meet the requirements for development of competitive solid-state batteries, which represents a major scientific challenge.[1-3]

Previous efforts to develop solid-state electrolytes have mainly focused on chalcogenides and oxides. Recently, metal hydrides have been discovered as a novel and promising class of solid-state ion conductors, *e.g.* based on large, low charge density anions. Fast sodium-ion conductivity of the order of 10^{-3} to 10^{-2} S·cm^{-1} at room temperature has been reported in *closo*-borate-based compounds and their derivatives,[4-6] while fast silver-ion conductivity in the order of 10^{-2} S·cm^{-1} is observed in $Ag_3B_{12}H_{12}I$.[7] The advantage of hydride-based solid electrolytes is their low reactivity with metal electrodes, low weight, and excellent deformability allowing for cheap assembly of the batteries, *e.g.* by cold pressing.[8] However, hydrides for Li-ion conductors usually have conductivities of 10^{-4} S·cm^{-1} or below at room temperature,[9-13] which is insufficient for utilization in all-solid-state batteries.

Recently, a high Li^+ conductivity of ~6.4×10^{-3} S·cm^{-1} at near room temperature, comparable to values of liquid organic electrolytes commonly employed in lithium-ion batteries, was reported in a lithium amide-borohydride, $Li(BH_4)_{0.33}(NH_2)_{0.67}$.[14] A half-cell battery with this solid-state electrolyte is stable over 400 cycles. However, the presence of a minor amount of an, until now, unidentified compound may contribute to the high Li^+ conductivity.[14] Herein, we report the discovery of hemiammine lithium borohydride, $LiBH_4·½NH_3$, with high Li-ion conductivity owing to ammonia-assisted Li^+ migration.

$LiBH_4·½NH_3$ was prepared by a mechanochemical reaction between $LiBH_4$, $LiNH_2$ and LiOH in the molar ratio 2:1:1, according to reaction 1.

$$2LiBH_4 + LiNH_2 + LiOH \rightarrow 2LiBH_4·½NH_3 + Li_2O \quad \text{(eq.1)}$$

Experimental details are provided in the *Electronic Supplementary Information*. The as-prepared sample (denoted **s1**) is characterized by synchrotron radiation powder X-ray diffraction (SR-PXD, Fig. S1) and the crystal structure of $LiBH_4·½NH_3$ was solved and Rietveld refined using this data and the software FOX and Fullprof.[15,16]

Monoammine lithium borohydride, $LiBH_4·NH_3$, was synthesized by a solid-gas reaction between $LiBH_4$ and NH_3 and also structurally investigated (Fig. S2).[17] The hemiammine compound, $LiBH_4·½NH_3$, was also synthesized with higher purity (94.8 wt%, see Fig. S3) by mechanical milling of $LiBH_4·NH_3$ and $LiBH_4$ in a molar ratio of 1:1, **s2**. The investigated samples are listed in Table 1.

Table 1. Investigated samples prepared by mechanochemistry.

Sample	Composition	Reactants
s1	66.9 wt% $LiBH_4·½NH_3$, 33.1 wt% Li_2O	$LiBH_4$–$LiNH_2$–LiOH (2:1:1)
s2	94.8 wt% $LiBH_4·½NH_3$, 5.2 wt% $LiBH_4$	$LiBH_4·NH_3$–$LiBH_4$ (1:1)
s3	15.1 wt% $LiBH_4·½NH_3$, 76.8 wt% $Li(BH_4)_{0.25}(NH_2)_{0.75}$, 8.1 wt% Li_2O	**s1** (25 wt%) – $Li(BH_4)_{0.25}(NH_2)_{0.75}$ (75 wt%)

[a] Interdisciplinary Nanoscience Center (iNANO) and Department of Chemistry, Aarhus University, Langelandsgade 140, 8000 Aarhus C, Denmark. Email: trj@chem.au.dk
[b] Institute of New Energy and Low-Carbon Technology, Sichuan University, Chengdu 610207, China.
[c] Center for Energy Materials Research, Korea Institute of Science and Technology, Seoul 02792, Republic of Korea
[d] Department of Quantum Matter Physics, Laboratory of Crystallography, University of Geneva, Quai Ernest-Ansermet 24, 1211 Geneva, Switzerland
Electronic Supplementary Information (ESI) available: [experimental details and additional experimental results]. See DOI: 10.1039/x0xx00000x

View Article Online
DOI: 10.1039/C9CC09990E

$LiBH_4·½NH_3$ crystallizes in an orthorhombic unit cell (space group *Pna*2_1) with unit cell parameters *a* = 9.6174(7), *b* = 13.7476(9) and *c* = 4.4409(5) Å, see Fig. 1A. Two unique Li positions are found in the structure (Fig. 1B). Both Li cations are tetrahedrally coordinated. One Li^+ is coordinated by two edge-sharing (κ^2) and two face-sharing (κ^3) BH_4^- with Li-B distances in the range 2.48-2.67 Å, while the second Li^+ is coordinated by two edge-sharing (κ^2) and one face-sharing (κ^3) BH_4^- and one NH_3 with Li-B distances of 2.46-2.66 Å and a Li-N bond distance of 1.97 Å. This results in coordination numbers (CN) of 10 and 8, respectively. The latter is similar to the reported coordination in the structure of $LiBH_4·NH_3$, in which all Li^+ are tetrahedrally coordinated by three BH_4^- and one NH_3 with Li-B distances of 2.52-2.57 Å and a Li-N bond distance of 2.01 Å.[17]

The structure of $LiBH_4·½NH_3$ is built by layers in the *a-c* plane formed by bridging BH_4^- groups. The layers, which are stacked along the *b*-axis, consist of alternating $[Li(BH_4)_4]$ and $[Li(BH_4)_3NH_3]$ tetrahedra along the *a*-axis, as shown in Fig. 1C. The layered structure may resemble that of the room temperature polymorph of $LiBH_4$, which is built by layers in the (*101*) plane interconnected to a framework *via* shared BH_4^- groups.[18] In contrast, the layers in $LiBH_4·½NH_3$ are interconnected by short dihydrogen bonds, 2.03-2.07 Å, between partly negatively charged $H^{\delta-}$ on BH_4^- and partly positively charged $H^{\delta+}$ on NH_3 with, $B–H^{\delta-} \cdots {}^{+\delta}H–N$. The dihydrogen bonds are slightly shorter than those in $LiBH_4·NH_3$.[17]

Fig. 2A shows the differential scanning calorimetry (DSC) profiles of **s1** in the first three heating/cooling cycles. An endothermic/exothermic peak is observed at 53 °C/45 °C in the heating/cooling process, indicating the occurrence of a first-order phase transition. Fig. 2B shows the *in-situ* synchrotron radiation powder X-ray diffraction (SR-PXD) pattern. All Bragg reflections corresponding to $LiBH_4·½NH_3$ disappear above 50 °C and a broad hump centered at 2θ = 14° emerges, indicating the molten state. The size of the Li_2O nano-particles is estimated to 9.2 nm using Rietveld refinement of the broad Bragg reflections and an isotropic size broadening model as implemented in the Fullprof program.

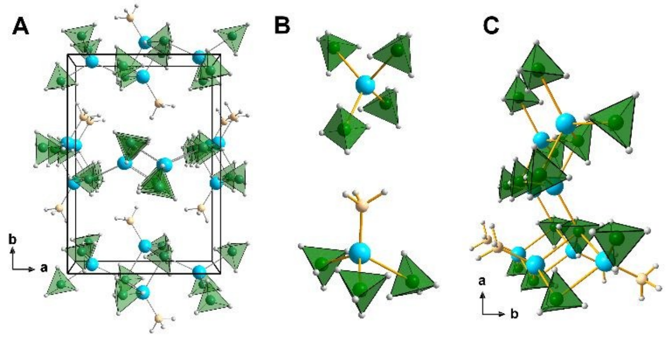

Figure 1. (A) Crystal structure of $LiBH_4·½NH_3$. (B) Two different Li coordination geometries. (C) The layer in the *a-c* plane. Color scheme: Li (light blue), N (yellow), H (grey) and BH_4 (green tetrahedrons).

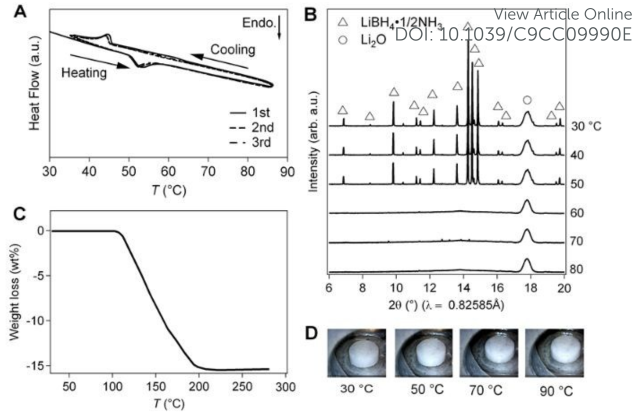

Figure 2. (A) Differential-scanning-calorimetry (DSC) profiles, (B) *in situ* synchrotron radiation powder X-ray diffraction (SR-PXD) patterns, (C) thermogravimetric (TG) data, and (D) temperature-programmed photographic analysis (TPPA) of sample **s1**.

The thermogravimetric (TG) measurement shows that the decomposition of **s1** starts at *T* ~ 100 °C (Fig. 2C). The observed mass loss of 15 wt% in the temperature range 110 to 220 °C, combined with mass spectrometry measurements, reveal that ammonia and hydrogen is released upon decomposition (Fig. S4). Interestingly, a pellet of **s1** maintained the shape when heated to 90 °C, *i.e.* above the melting point (Fig. 2D), revealing a mechanical stabilization of the molten state by inert nano-particles. The pellet shape of sample **s1** is retained up to 125 °C (Fig. S5), above which the pellet deforms and expands, due to the gas release as a result of the decomposition of $LiBH_4·½NH_3$.

In contrast, the pellet of sample **s2** collapsed at 60 °C (Fig. S6), indicating the decomposition of sample **s2**. Heating **s2** at 60 °C in vacuum leads to the formation of $LiBH_4$ (Fig. S7). The improved thermal stability in sample **s1** over sample **s2** is attributed to the presence of nano-crystalline Li_2O.

The Li^+ conduction mechanism in $LiBH_4·½NH_3$ has been investigated by density functional theory (DFT) calculations by introducing an interstitial Li^+-ion in the crystal structure. The energy profile of the interstitial Li^+ migrating through the unit cell, along the *c*-axis in between the layers in the *a-c* plane is shown in Fig. 3. The energy barrier is calculated to be as small as 0.16 eV. In line with results of DFT calculations, bond-valence-based empirical force field calculations using SoftBV program also indicate a 1D conduction pathway along the *c*-axis (Figs. S8 and S9). The coordination of the interstitial Li^+ is illustrated in Fig. 3, C0 to C2. Surprisingly, the interstitial Li^+ 'borrows' an ammonia molecule from the framework Li^+. After a jump forward, the interstitial Li^+ deliver the NH_3 molecule back to the framework Li^+. To facilitate this exchange the framework Li^+ is displaced up to 1.14 Å into a trigonal coordination geometry by three BH_4^- groups.

View Article Online
DOI: 10.1039/C9CC09990E

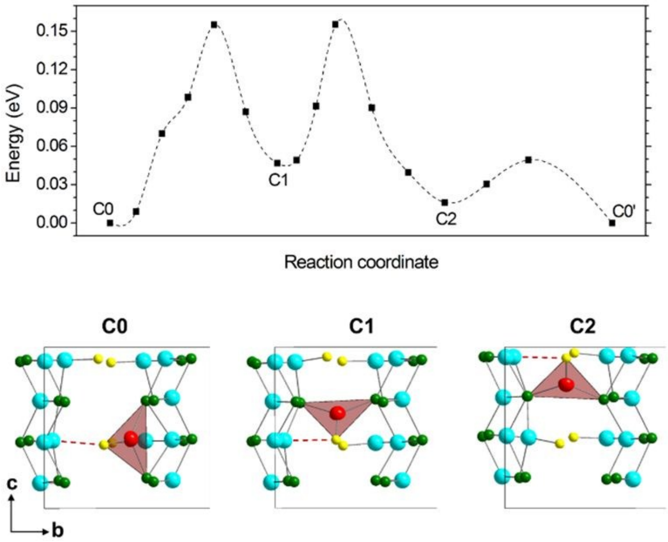

Figure 3. The energy profile and the coordination (denoted C0 to C2) of an interstitial Li^+ ion migrating along the c-axis in the interlayer space in the a-c plane of the $LiBH_4{\cdot}½NH_3$ structure. Framework Li^+ is shown as blue spheres and interstitial Li^+ as red, N is yellow and B is green. H atoms are omitted for clarity. The dotted red line illustrates the temporarily broken bond between framework Li and NH_3.

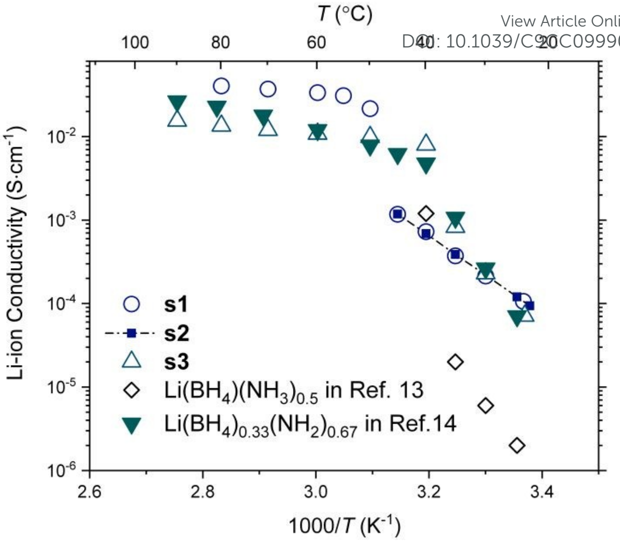

Figure 4. Temperature-dependent Li-ion conductivities of samples **s1**, **s2** and **s3,** compared to that of $LiBH_4(NH_3)_{0.5}$ reported in Ref. [13] and $Li(BH_4)_{0.33}(NH_2)_{0.67}$ reported in Ref. [14].

The interstitial Li^+ is initially (C0) coordinated in a trigonal geometry by two BH_4^- groups (κ^2) and one NH_3, while an additional BH_4^- coordinates through one H (κ^1). The displaced framework Li^+ is stabilized by a reorientation of two BH_4^- group, changing from edge (κ^2) to face-sharing (κ^3) coordination of the BH_4, to maintain the coordination number 8 for Li^+. As the interstitial Li^+ moves from C0 to C1, it remains bonded to the same NH_3, while exchanging the coordination from one BH_4^- complex to another BH_4^- in the adjacent layer. In C1, the interstitial Li^+ is coordinated in a trigonal geometry by one NH_3 and two BH_4^- groups through the edge and the face of the tetrahedra (κ^2, κ^3), while the framework Li^+ is coordinated by two edge-sharing (κ^2) and one face-sharing (κ^3) BH_4^- complex. Moving from C1 to C2, the bond between interstitial Li^+ and the NH_3 is broken, but is reformed with another NH_3 molecule, while keeping the coordination to the two BH_4^- groups (κ^2, κ^3). The NH_3 molecule again coordinates to the original framework Li^+ ion and has effectively assisted the migration of interstitial Li^+. In this process nitrogen is displaced up to 0.67 Å (and NH_3 is rotated) as it changes coordination between framework and interstitial Li^+. Moving from C2 to C0, the bond to the NH_3 is preserved, while one of the BH_4^- complexes is exchanged.

In other words, each movement from/to a local minimum involves an exchange of either one NH_3 or BH_4^- group, and the overall process gives a net displacement of only the interstitial Li^+ along the c-direction, without involving the net migration of other units. The interstitial Li^+ maintains CN = 6 during migration, while framework Li^+ has CN = 8 in C0 and C2 and CN = 7 in C1. Although we did not explore other migration mechanisms, the proposed interstitial mechanism with a small energy barrier corroborates the fast Li^+ conduction in $LiBH_4{\cdot}½NH_3$.

The temperature-dependent Li^+ conductivities of the investigated samples are shown in Fig. 4. Sample **s1** exhibits a Li-ion conductivity of $\sigma(Li^+) = 1 \times 10^{-4}$ S·cm^{-1} at 25 °C, which is approximately four orders of magnitude larger than that of $LiBH_4$. Upon heating, $\sigma(Li^+)$ further increases to 3×10^{-2} S·cm^{-1} at 55 °C owing to the melting of $LiBH_4{\cdot}½NH_3$.

The activation energy estimated from the slope of ln(σT) versus $1/T$ (see Table S2) is $E_a = 0.97$ eV at $T < 55$ °C and $E_a = 0.13$ eV at $T > 55$ °C. Sample **s2** shows a similar conductivity to that of **s1** within the temperature range 25 to 45 °C. The electrochemical stability window of $LiBH_4{\cdot}½NH_3$ was determined to be 0.2 – 1.8 V *vs* Li/Li^+ (Figure S10).

The Li^+ conductivity reported recently for ammine lithium borohydride, $LiBH_4(NH_3)_x$ ($x = 0 - 1$),[13] is similar to that of **s1** at 40 °C. However, at lower temperatures, **s1** has a significantly higher conductivity of $\sigma(Li^+) = 1.0 \times 10^{-4}$ S·cm^{-1} at 25 °C as compared to 2.0×10^{-6} S·cm^{-1} observed for $LiBH_4(NH_3)_{0.5}$. Careful analysis of the PXD pattern of the $LiBH_4(NH_3)_{0.5}$ sample in Ref.[13] suggests that this sample is in fact a mixture of $LiBH_4$, $LiBH_4{\cdot}NH_3$ and possibly a minor amount of $LiBH_4{\cdot}½NH_3$.

To investigate the influence of Li_2O in the samples, different amounts of Li_2O were introduced to sample **s1** (Figure S11, Table S1). Addition of up to 75 wt% Li_2O does not improve the conductivities at room temperature significantly. However, addition of 80 wt% Li_2O markedly reduces the conductivity to 1.1×10^{-5} S·cm^{-1} at 25 °C.

Additionally, we found that $LiBH_4{\cdot}½NH_3$ has appeared as a minor byproduct in the synthesis of lithium amide-borohydride, $Li(BH_4)_{1-x}(NH_2)_x$ (x = 0 – 1), and has previously, mistakenly, been assigned as γ–$Li(BH_4)_{1-x}(NH_2)_x$.[14, 19] Here we have re-analyzed X-ray data of the $Li(BH_4)_{0.33}(NH_2)_{0.67}$ sample reported in Ref. [14], which in fact is composed of the cubic polymorph α-$Li(BH_4)_{0.25}(NH_2)_{0.75}$ (~85 wt%) and $LiBH_4{\cdot}½NH_3$ (~15 wt%) (Fig. S12). It is known that $Li(BH_4)_{0.25}(NH_2)_{0.75}$ maintains a cubic polymorph until the decomposition at $T > 250$ °C and shows $\sigma(Li^+) \sim 1 \times 10^{-4}$ S·cm^{-1} at 40 °C.[20] Thereby, the reported first-order transition near 40 °C and the high $\sigma(Li^+)$ of 6.4×10^{-3} S·cm^{-1} at 40 °C of $Li(BH_4)_{0.33}(NH_2)_{0.67}$ (see Ref. [14]) can be attributed to the presence of ~ 15.0 wt% $LiBH_4{\cdot}½NH_3$. The transition temperature at 40 °C observed in $Li(BH_4)_{0.33}(NH_2)_{0.67}$ is approximately 15 °C lower than that of $LiBH_4{\cdot}½NH_3$, which could be attributed to an interface effect between $LiBH_4·½NH_3$ and α-$Li(BH_4)_{0.25}(NH_2)_{0.75}$.

A sample consisting of **s1** (25 wt%) and α-$Li(BH_4)_{0.25}(NH_2)_{0.75}$ (75 wt%) prepared by mechanical milling (denoted **s3**), which contains 15.1 wt% $LiBH_4·½NH_3$ (Figs. S13-14) was also investigated. Fig. 4 reveals that the temperature-dependent Li^+ conductivity of **s3** resembles that of $Li(BH_4)_{0.33}(NH_2)_{0.67}$, confirming that the high conductivity of $Li(BH_4)_{0.33}(NH_2)_{0.67}$ reported in Ref. [14] is in fact owing to the presence of a minor amount of $LiBH_4·½NH_3$. In other words, the $LiBH_4·½NH_3$ melt can be mechanically stabilized with other materials such as $Li(BH_4)_{0.25}(NH_2)_{0.75}$ (Fig. S15). The potential as a solid-state electrolyte is evidenced by the stable cycling of $Li(BH_4)_{0.33}(NH_2)_{0.67}$ in a half-cell battery over 400 cycles.[11] Here we show that the $LiBH_4·½NH_3$ melt can also be mechanically stabilized by metal oxides (such as Li_2O, Fig. S16), which may form better contact with the electrodes.

In summary, we report a new hemiammine lithium borohydride, $LiBH_4·½NH_3$, exhibiting a Li-ion conductivity of 7×10^{-4} S·cm^{-1} at 40 °C, which increases to 3.0×10^{-2} S·cm^{-1} at 55 °C as a result of melting. DFT calculations reveal a new mechanism for cation migration where NH_3 stabilizes an interstitial Li^+ cation and facilitates fast Li-ion migration in the *a-c* plane in the structure of $LiBH_4·½NH_3$. We also identify that the new compound reported in this investigation, $LiBH_4·½NH_3$, occurs as a minor component in the $Li(BH_4)_{1-x}(NH_2)_x$ system (see Refs. [14] and [19]) and is contributing to the high Li^+ conductivity. Although $LiBH_4·½NH_3$ melts at $T \sim 50$ °C, combination with inert materials such as Li_2O, can be an effective approach to create a mechanically stable material. Further efforts may stabilize the highly conductive state observed at $T > 50$ °C down to or below room temperature. This illustrates that solid state ion transport in a composite material is a multiscale process,[3] where atomic scale cation migration occur through the solid crystalline matter, *i.e.* via the new ammonia assisted cation migration mechanism presented here, which may be further enhanced by grain-boundary diffusion on inert, insulating nano-particles.[10] Here, a molten phase was also stabilized by insulating nano-particles to form a mechanically stable material with high Li^+ conductivity. These effects may be further optimized to develop technologically useful electrolytes for future solid-state batteries.

This work was supported by the Danish council for independent research, technology and production (HyNanoBorN, 4181-00462 and SOS-MagBat, 9041-00226B). NordForsk and The Nordic Neutron Science Program are acknowledged for funding (FunHy, project no. 81942), and Sichuan University for a start-up talent recruitment project (YJ201914) and National Natural Science Foundation of China (21975168). The work at Korea Institute of Science and Technology (KIST) was supported by the KIST Institutional Program (Project No. 2E29630). The authors would like to thank the Diamond Light Source for access to the I11 beamline and the local contacts Dr. Stephen Thompson and Dr. Sarah Day for assistance with data collection.

Conflicts of interest

There are no conflicts to declare.

Notes and references

View Article Online
DOI: 10.1039/C9CC09990E

CCDC 1973062 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

1. H. Zhang, C. M. Li, M. Piszcz, E. Coya, T. Rojo, L. M. Rodriguez-Martinez, M. Armand and Z. B. Zhou, *Chem. Soc. Rev.* 2017, **46**, 797-815.
2. K. H. Park, Q. Bai, D. Hyeon Kim, D. Y. Oh, Y. Zhu, Y. Mo and Y. S. Jung, *Adv. Energy. Mater.* 2018, **8**, 1800035.
3. T. Famprikis, P. Canepa, J. A. Dawson, M. S. Islam and C. Masquelier, *Nat. Mater.* 2019, **18**, 1278-1291.
4. Y. Sadikin, M. Brighi, P. Schouwink and R. Černý, *Adv. Energy. Mater.* 2015, **5**, 1501016.
5. L. Duchene, R. S. Kuhnel, D. Rentsch, A. Remhof, H. Hagemann and C. Battaglia, *Chem. Commun.* 2017, **53**, 4195-4198.
6. W. S. Tang, K. Yoshida, A. V. Soloninin, R. V. Skoryunov, O. A. Babanova, A. V. Skripov, M. Dimitrievska, V. Stavila, S. Orimo and T. J. Udovic, *ACS Energy Lett.* 2016, **1**, 659-664.
7. M. Paskevicius, B. R. S. Hansen, M. Jørgensen, B. Richter and T. R. Jensen, *Nat. Commun.* 2017, **8**, 15136.
8. J. Cuan, Y. Zhou, T. Zhou, S. Ling, K. Rui, Z. Guo, H. Liu and X. Yu, *Adv. Mater.* 2019, **31**, 1803533.
9. F. Lu, Y. Pang, M. Zhu, F. Han, J. Yang, F. Fang, D. Sun, S. Zheng and C. Wang, *Adv. Funct. Mater.* 2019, **0**, 1809219.
10. Y. S. Choi, Y.-S. Lee, D.-J. Choi, K. H. Chae, K. H. Oh and Y. W. Cho, *J. Phys. Chem. C*, 2017, **121**, 26209-26215.
11. D. Blanchard, A. Nale, D. Sveinbjornsson, T. M. Eggenhuisen, M. H. W. Verkuijlen, Suwarno, T. Vegge, A. P. M. Kentgens and P. E. de Jongh, *Adv. Funct. Mater.* 2015, **25**, 184-192.
12. K. T. Møller, M. Paskevicius, J. G. Andreasen, J. Lee, N. Chen-Tan, J. Overgaard, S. Payandeh, D. S. Silvester, C. E. Buckley and T. R. Jensen, *Chem. Commun.* 2019, **55**, 3410-3413.
13. T. Zhang, Y. Wang, T. Song, H. Miyaoka, K. Shinzato, H. Miyaoka, T. Ichikawa, S. Shi, X. Zhang, S. Isobe, N. Hashimoto and Y. Kojima, *Joule*, 2018, **2**, 1522-1533.
14. Y. Yan, R.-S. Kühnel, A. Remhof, L. Duchêne, E. C. Reyes, D. Rentsch, Z. Łodziana and C. Battaglia, *Adv. Energy Mater.* 2017, **7**, 1700294.
15. V. Favre-Nicolin and R. Cerny, *J Appl Crystallogr*, 2002, **35**, 734-743.
16. J. Rodriguez-Carvajal, Fullprof, *LLB Sacley & LCSIM Rennes*: France, 2003.
17. S. R. Johnson, W. I. F. David, D. M. Royse, M. Sommariva, C. Y. Tang, F. P. A. Fabbiani, M. O. Jones and P. P. Edwards, *Chem. Asian J.* 2009, **4**, 849-854.
18. J. P. Soulié, G. Renaudin, R. Černý and K. Yvon, *J. Alloy. Compd.* 2002, **346**, 200-205.
19. G. P. Meisner, M. L. Scullin, M. P. Balogh, F. E. Pinkerton and M. S. Meyer, *J. Phys. Chem. B*, 2006, **110**, 4186-4192.
20. M. Matsuo, A. Remhof, P. Martelli, R. Caputo, M. Ernst, Y. Miura, T. Sato, H. Oguchi, H. Maekawa, H. Takamura, A. Borgschulte, A. Zuttel and S. Orimo, *J. Am. Chem. Soc.* 2009, **131**, 16389-16391.